



विद्यया न सर्वं धनं प्राप्यते

IIT JAMMU

IIT Jammu

MTech in GenAI and Data Science

Semester 1 | Course Curriculum

Semester 1 — Course Overview

Type	Course	Faculty	Credits
Core	Programming for AI and Data Science	Dr. Vinit Jakhetiya	3
Core	Data Analytics for AI	Dr. Badri	3
Core	Foundation for Machine Learning and Deep Learning	Dr. Soma Dhavala	3
Core	Artificial Intelligence and Computational Intelligence	Dr. Ashutosh Yadav and Dr. Karan Nathwani	3

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Programming for AI and Data Science

Credits: 3

L-D-P Structure:	3-0-0
Credits:	3

Course Objectives

Python is one of the most versatile and widely used scripting languages. Its clean and uncluttered syntax as well as its straightforward design greatly contribute to this success and make it an ideal language for programming education. Its application ranges from web development to scientific computing. Especially in the fields of data science and artificial intelligence, it is the most common programming language supported by all major data-handling and analytical frameworks. This course provides a thorough introduction to the language and its main features, as well as insights into the rationale and application of important adjacent concepts such as environments, testing, and version control. Also in this course, students will learn how to code for building data science / machine learning applications.

Course Outcomes

- Understand Python syntax and programming concepts.
- Understand object-oriented concepts in Python.
- Analyze and apply different methods for error handling in Python.
- Understand important Python libraries related to data science and how to apply them to given programming tasks.
- Understand concepts like environments and version control.

Course Contents

- Introduction to Data Science using Python
- Python basic constructs
- Object Oriented Programming in Python
- NumPy for mathematical computing
- SciPy for scientific computing
- Data manipulation
- Data visualization with Matplotlib
- Machine learning libraries in Python — Pandas, Matplotlib, NumPy, scikit-Learn

Detailed Course Contents

No.	Topic & Contents	Hours
1	Introduction to Data Science using Python What is Data Science, what does a data scientist do. Various examples of Data Science in the industries. How Python is deployed for Data Science applications. Various steps in the Data Science process like data wrangling, data exploration and selecting the model. Introduction to Python programming language. Important Python features, how is Python different from other programming languages. Python installation, Anaconda Python distribution for Windows, Linux and Mac. How to run a sample Python script, Python IDE working mechanism. Running some Python basic commands. Python variables, data types and keywords.	6

No.	Topic & Contents	Hours
2	Python Basic Constructs Introduction to a basic construct in Python. Understanding indentation like tabs and spaces. Python built-in data types. Basic operators in Python. Loop and control statements like break, if, for, continue, else, range() and more. File Handling (I/O) and Exception Handling, Pandas.	10
3	Object Oriented Programming in Python Understanding the OOP paradigm like encapsulation, inheritance, polymorphism and abstraction. What are access modifiers, instances, class members. Classes and objects. Function parameter and return type functions. Lambda expressions.	8
4	NumPy for Mathematical Computing Introduction to mathematical computing in Python. What are arrays and matrices, array indexing, array math, Inspecting a NumPy array, NumPy array manipulation.	6
5	SciPy for Scientific Computing Introduction to SciPy. Functions building on top of NumPy, cluster, linalg, signal, optimize, integrate, subpackages, SciPy with Bayes Theorem.	6
6	Data Manipulation Loading data from various files (.dat, .json, .h5, .txt, .csv, .xlsx etc.). Example applications.	3
7	Data Visualization with Matplotlib Introduction to Matplotlib. Using Matplotlib for plotting graphs and charts like Scatter, Bar, Pie, Line etc. Histogram and more Matplotlib APIs.	3
Total Lecture Hours		42

Suggested Texts & Reference Materials

1. Wes McKinney: *Python for Data Analysis: Data Wrangling with Pandas, NumPy, and IPython*, O'Reilly Media, 2017. Supplementary reading material and other resources may be provided in class throughout the semester.

Data Analytics for AI

Credits: 3

L-D-P Structure:	3-0-0
Credits:	3

Course Objectives

The objective of the course is to provide the fundamental knowledge of probability and random theory with an application to artificial intelligence and machine learning.

Course Outcomes

- Understand the concepts of random variables and random processes.
- Comprehend the various statistical models used for modeling for machine learning application.
- Apply the different detection and estimation rules in real time problems.

Course Contents

- Fundamentals of Probability and Random Signal Theory
- Random Processes and Convergence
- Markov Chains and Applications
- Frequentist Statistics and Regression
- Bayesian Statistics and Hypothesis Testing

Detailed Course Contents

No.	Topic & Contents	Hours
1	Fundamentals of Probability and Random Signal Theory Probability spaces, Conditional probability, Independence, Random Variables, Discrete and Continuous random variables, Expectation Operator, Functions of random variables, Generating random variables, Multivariate Random Variables, Joint distributions of discrete and continuous variables, Functions of several random variables, Joint Moments, Generating multivariate random variables.	9
2	Random Processes and Convergence Definition, Stationarity of random processes, Mean and autocovariance functions, Independent identically-distributed sequences, Power spectral density, Gaussian process, Poisson process, Random walk, Convergence of Random Processes, Types of convergence, Law of large numbers, Central limit theorem, Monte Carlo simulation.	9
3	Markov Chains and Applications Time-homogeneous discrete-time Markov chains, Recurrence, Periodicity, Convergence, Markov-chain Monte Carlo, Descriptive statistics, Histogram, Sample mean and variance, Order statistics, Sample covariance, Sample covariance matrix.	8
4	Frequentist Statistics and Regression Independent identically-distributed sampling, Mean square error, Consistency, Confidence intervals, Nonparametric model estimation, Parametric model estimation, Linear Regression models, Least-squares estimation, Overfitting, Global warming.	8
5	Bayesian Statistics and Hypothesis Testing Bayesian parametric models, Conjugate prior, Bayesian estimators, The hypothesis-testing framework, Parametric testing, Nonparametric testing: The permutation test, Multiple testing.	8

Total Lecture Hours	42
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Suggested Texts & Reference Materials

1. Carlos Fernandez-Granda: *Probability and Statistics for Data Science*.
 2. Anirban Das Gupta: *Probability for Statistics and Machine Learning: Fundamentals and Advanced Topics*.
 3. Athanasios Papoulis: *Probability, Random Variables, and Stochastic Processes*.
- Supplementary reading material and other resources may be provided in class throughout the semester.



Artificial Intelligence and Computational Intelligence

Credits: 3

L-D-P Structure:	3-0-0
Credits:	3

Course Objectives

This course introduces the foundational principles and techniques of Artificial Intelligence (AI) and Computational Intelligence (CI). Students will learn how intelligent agents perceive, reason, learn, and act. The course explores classical AI approaches such as search, logic, reasoning, and knowledge representation, as well as computational intelligence approaches like neural networks, fuzzy logic, and evolutionary algorithms. The goal is to equip students with the ability to model intelligent behaviour and apply AI/CI techniques to real-world problem-solving.

Course Outcomes

- Explain the fundamental concepts of Artificial Intelligence and Computational Intelligence.
- Model problems using intelligent agents, search strategies, and knowledge representation techniques.
- Apply classical AI approaches (search, logic, reasoning) to solve real-world problems.
- Implement computational intelligence methods such as neural networks, fuzzy systems, and evolutionary algorithms.
- Analyze and compare different AI/CI techniques for specific domains and applications.
- Build small-scale intelligent systems using appropriate AI/CI methodologies.

Course Contents

- Introduction to AI and Intelligent Agents
- Problem Formulation & Search Strategies
- Knowledge Representation & Reasoning
- Logical Inference & Planning
- Machine Learning Fundamentals
- Neural Networks (Introduction)
- Fuzzy Logic and Approximate Reasoning
- Evolutionary Computing (Genetic Algorithms)
- Swarm Intelligence
- Applications of AI/CI in real-world systems

Detailed Course Contents

No.	Topic & Contents	Hours
1	Introduction to AI and Intelligent Agents AI problems, foundation of AI and history of AI. Intelligent agents: Agents and Environments, the concept of rationality, the nature of environments, structure of agents, problem solving agents, problem formulation.	6
2	Search Strategies Searching for solutions, uninformed search strategies — Breadth first search, Depth first search. Search with partial information (Heuristic search): Hill climbing, A*, AO* Algorithms, Problem reduction. Game Playing — Adversarial search, Games, Mini-max algorithm, Optimal decisions in multiplayer games, Alpha-Beta pruning, Evaluation functions.	8

No.	Topic & Contents	Hours
3	Knowledge Representation & Reasoning Knowledge representation issues, predicate logic, logic programming, semantic nets, frames and inheritance, constraint propagation, representing knowledge using rules, rules-based deduction systems. Reasoning under uncertainty, review of probability, Bayes probabilistic inferences and Dempster-Shafer theory.	8
4	Logical Inference & Planning First order logic. Inference in first order logic, propositional vs. first order inference, unification & lifts, forward chaining, backward chaining, resolution. Learning from observation, inductive learning, decision trees, explanation-based learning, statistical learning methods, reinforcement learning.	8
5	Expert Systems & Applied AI Introduction, basic concepts, structure of expert systems, the human element in expert systems. How expert systems work, problem areas addressed by expert systems, expert systems success factors, types of expert systems. Knowledge engineering, scope of knowledge, difficulties in knowledge acquisition, methods of knowledge acquisition, machine learning, intelligent agents. Societal impacts, reasoning in AI, inference with rules, with frames, model-based reasoning, case-based reasoning, explanation & meta knowledge, inference with uncertainty.	10
6	Applications of AI	2
Total Lecture Hours		42

Suggested Texts & Reference Materials

1. S. Russel and P. Norvig: *Artificial Intelligence – A Modern Approach*, Second Edition, Pearson Education.
2. David Poole, Alan Mackworth, Randy Goebel: *Computational Intelligence: A Logical Approach*, Oxford University Press.
3. G. Luger: *Artificial Intelligence: Structures and Strategies for Complex Problem Solving*, Fourth Edition, Pearson Education.
4. J. Nilsson: *Artificial Intelligence: A New Synthesis*, Elsevier Publishers.

Supplementary reading material and other resources may be provided in class throughout the semester.

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Foundation for Machine Learning and Deep Learning

Credits: 3

L-D-P Structure:	3-0-0
Credits:	3

Course Objectives

Machine learning is a field of scientific study concerned with algorithmic techniques that enable machines to learn performance on a given task via the discovery of patterns or regularities in exemplary data. Consequently, its methods commonly draw upon a statistical basis in conjunction with the computational capabilities of modern computing hardware. This course aims to acquaint the student with the main branches of machine learning and provide a thorough introduction to the most widely used approaches and methods in this field.

Course Outcomes

- Know different machine learning model classes.
- Comprehend the difference between supervised, unsupervised, and reinforcement learning methods.
- Understand common machine learning models.
- Analyze trade-offs in the application of different models.
- Appropriately choose machine learning models according to a given task.

Course Contents

- Introduction to Machine Learning
- Clustering
- Regression
- Support Vector Machines
- Decision Trees
- Genetic Algorithms

Detailed Course Contents

No.	Topic & Contents	Hours
1	Introduction to Machine Learning Learning Hypothesis. Data Representation and Pre-processing. Types of Learning: Supervised, Unsupervised, Semi-supervised, Reinforcement Learning. Regression, Classification, Clustering. Metrics for accuracy: Precision, Recall, F1-score, total accuracy, MSE, MAPE etc. Data Split and Data Pre-processing, cross-validation and parameter tuning.	6
2	Challenges in Learning PAC-learning. Bias and Variance. Data Representation and Pre-processing.	4
3	Supervised Learning Linear Regression, Non-Linear Regression, Regularization, Quantile and multivariate. Optimization (Gradient Descent) and Closed form solutions. Logistic Regression (LogReg), LogReg with Regularization. Bayesian and Naive Bayes Classification. Support Vector Machine (linear, non-linear kernel) and Support Vector Regression, Various optimisation schemes (SMO, GD etc). Decision Tree, Random Forest, Neural Network.	20
4	Unsupervised Clustering Introduction to clustering. K-Means. Expectation Maximization. DBScan. Hierarchical Clustering.	10

No.	Topic & Contents	Hours
5	Hyper-parameter Optimisation Grid Search. Random Search.	2
Total Lecture Hours		42

Suggested Texts & Reference Materials

1. D.W. Patterson: *Introduction to AI and Expert Systems*, McGraw-Hill, 1998.
 2. C. Bishop: *Pattern Recognition and Machine Learning*, Springer-Verlag, 2006.
- Supplementary reading material and other resources may be provided in class throughout the semester.

